PVMol-Gen Replication Pipeline

Abstract Workflow with Tool Requirements

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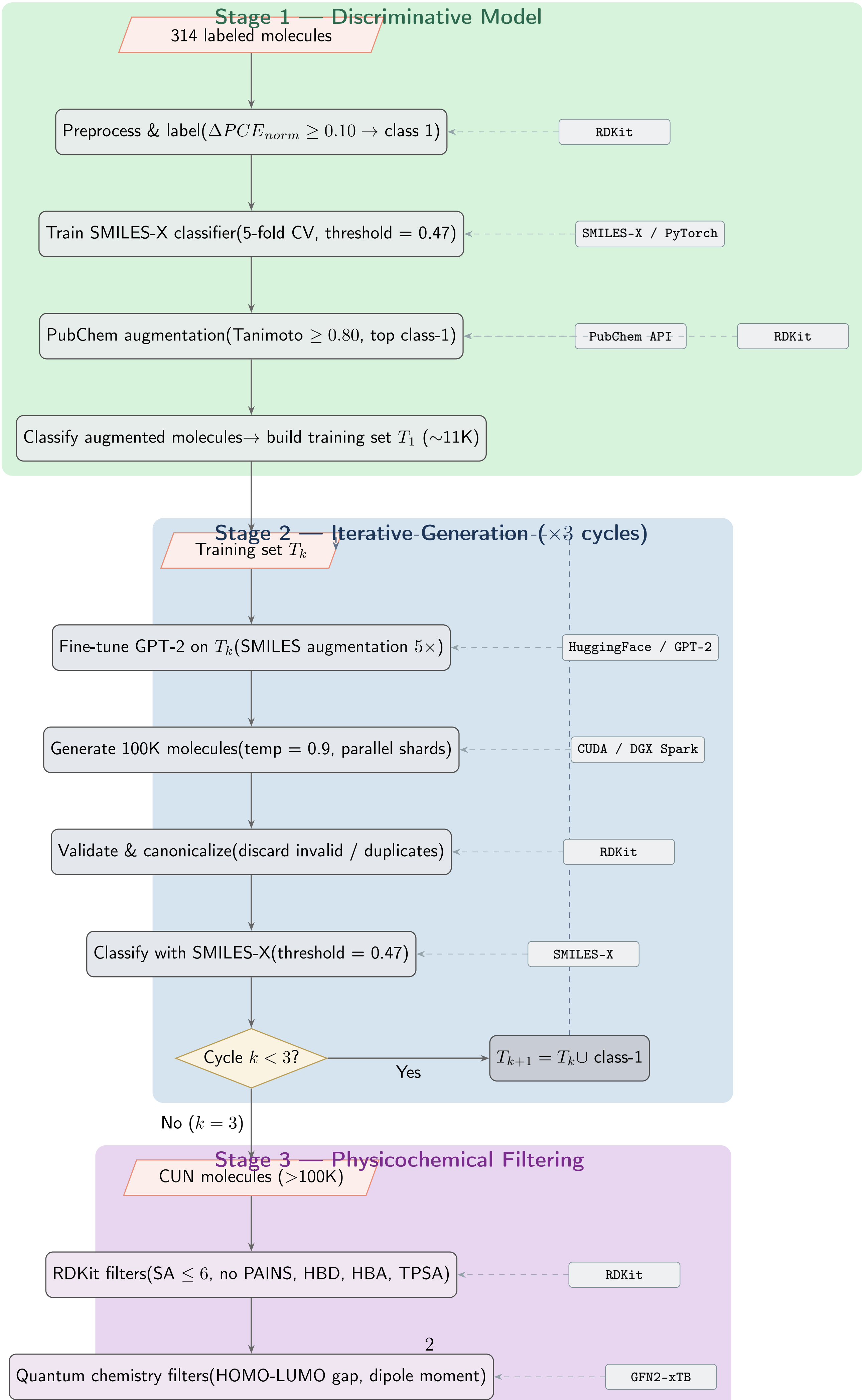
Pipeline Overview


Stage 1 — Discriminative Model
314 labeled molecules
Preprocess & label($\Delta PCE_{norm} \geq 0.10 \rightarrow$ class 1)
RDKit
Train SMILES-X classifier(5-fold CV, threshold = 0.47)
SMILES-X / PyTorch
PubChem augmentation(Tanimoto ≥ 0.80, top class-1)
PubChem API
RDKit
Classify augmented molecules$\rightarrow$ build training set T_1 (∼11K)
Stage 2 — Iterative Generation ($\times 3$ cycles)
Training set T_k
Fine-tune GPT-2 on T_k(SMILES augmentation $5\times$)
HuggingFace / GPT-2
Generate 100K molecules(temp = 0.9, parallel shards)
CUDA / DGX Spark
Validate & canonicalize(discard invalid / duplicates)
RDKit
Classify with SMILES-X(threshold = 0.47)
SMILES-X
Cycle $k < 3$?
Yes
$T_{k+1} = T_k \cup$ class-1
No ($k = 3$)
Stage 3 — Physicochemical Filtering
CUN molecules (>100K)
RDKit filters(SA ≤ 6, no PAINS, HBD, HBA, TPSA)
RDKit
Quantum chemistry filters(HOMO-LUMO gap, dipole moment)
GFN2-xTB

Tool & Dependency Matrix

Software Stack Summary

Key Decision Points

1. **Classification threshold** — Optimized to 0.47 (not default 0.5) to maximize F1 on validation data. This shifts the class-1 boundary and propagates through all downstream steps.
2. **Number of cycles** — Fixed at 3. More cycles risk model collapse (generative model trained on its own increasingly synthetic data).
3. **Generation temperature** — $T = 0.9$ balances diversity and validity. Lower T yields more valid but less diverse molecules; higher T increases novelty at the cost of validity rate.
4. **Parallel sharding** — Required for practical wall-clock time. Single-machine generation of 100K valid molecules takes $\sim$19 hours; 3-way parallelism reduces this to $\sim$3 hours.
5. **xTB vs. DFT** — Semi-empirical GFN2-xTB is used for energy gap and dipole filters (minutes per molecule vs. hours for full DFT). Sufficient accuracy for screening; DFT reserved for final candidates.
6. **Retrain from scratch** — GPT-2 is retrained from the pretrained checkpoint each cycle (not continued from previous fine-tune), ensuring the model learns the full expanded distribution.

Step	Operation	Tools / Libraries	Compute
Stage 1 — Discriminative Model			
1.1	Assemble 314-molecule dataset	Pandas, RDKit (SMILES validation)	CPU
1.2	Compute labels (ΔPCE_{norm})	NumPy	CPU
1.3	Train SMILES-X (5-fold CV)	PyTorch, SMILES-X framework	GPU (DGX Spark)
1.4	Optimize threshold ($\rightarrow 0.47$)	scikit-learn (F1 optimization)	CPU
1.5	PubChem similarity search	PubChem REST API, RDKit (Morgan FP)	CPU + network
1.6	Classify augmented $\rightarrow$ build T_1	Trained SMILES-X model	GPU
Stage 2 — Iterative Generation ($\times$3 cycles)			
2.1	SMILES augmentation (5$\times$)	RDKit (random SMILES)	CPU
2.2	Fine-tune GPT-2 on T_k	HuggingFace Transformers, Accelerate	GPU (DGX Spark)
2.3	Generate 100K molecules	GPT-2 (autoregressive, $T=0.9$)	3$\times$ GPU (parallel shards)
2.4	Validate & canonicalize	RDKit (MolFromSmiles, canonical)	CPU
2.5	Classify $\rightarrow$ class-1 set	Trained SMILES-X model	GPU
2.6	Expand training set T_{k+1}	Pandas (merge & deduplicate)	CPU
Stage 3 — Physicochemical Filtering			
3.1	SA score ≤ 6	RDKit (sa_score)	CPU
3.2	PAINS filter	RDKit (FilterCatalog)	CPU
3.3	HBD $\in [0, 2]$, HBA $\in [2, 5]$	RDKit (Descriptors)	CPU
3.4	TPSA $\in [50, 120]$ Å^2	RDKit (Descriptors.TPSA)	CPU
3.5	3D conformer generation	RDKit (ETKDG)	CPU
3.6	HOMO-LUMO gap $\in [1.5, 5.0]$ eV	GFN2-xTB (geometry opt + orbitals)	CPU (batch)
3.7	Dipole moment $\in [1.5, 4.0]$ D	GFN2-xTB (from opt output)	CPU (batch)
3.8	Agglomerative clustering (10 groups)	scikit-learn, RDKit (Morgan FP)	CPU
3.9	Sample 1 representative / cluster	NumPy (random choice)	CPU

Table 1: Full step-by-step tool and compute requirements.

Component	**Version**	**Purpose**
Python	3.11	Runtime
PyTorch	≥ 2.4	Neural network training (SMILES-X, GPT-2)
HuggingFace Transformers	≥ 5.0	GPT-2 fine-tuning & generation
Accelerate	≥ 1.0	Distributed / mixed-precision training
RDKit	latest (conda-forge)	Molecular operations (validation, FP, descriptors)
scikit-learn	≥ 1.5	Clustering, threshold optimization
GFN2-xTB	≥ 6.6	Semi-empirical quantum chemistry
Pandas / NumPy	latest	Data wrangling
PubChem PUG-REST	(web API)	Similarity search for data augmentation
Miniforge	latest	Environment management across machines

Table 2: Software dependencies.